Information Security

Muhammad Bilal, Ph.D.

Ph.D. Computer Science and Technology, Zhejiang University, China

MS Computer Science, NTU Faisalabad Pakistan

B.Sc (Hons) Computer Science, UET Lahore Pakistan

GC University Faisalabad

Motivation


فَبِاَيِّ اٰلَآءِ رَبِّكُمَا تُكَذِّبٰنِ
تو (اے جن و انسان!) تم دونوں اپنے رب کی کون کون سی نعمتوں کو جھٹلاؤ گے؟
SURAH REHMAN 55:13

Why Work with Security?

- Main aim of security is to **protect valuable assets of a company or individual.**
- Involved activities such as:
  - Ensuring unauthorized people do not have access to assets
  - Ensuring authorized people do have access to assets
  - Preventing loss of assets by theft or physical damage
- To do the job well we must:
  - Decide who is **allowed** to do what
  - Analyze **risks** which may threaten assets
  - Consider **controls and countermeasures**

Threats and Vulnerabilities

- **Vulnerability:** A weakness which can be exploited to cause loss or harm.
- **Threat:** A set of circumstances that has the potential to cause loss or harm.



Here is a picture of a threat.

Can you see the vulnerability?

A threat is blocked by control of a vulnerability.

Examples of Threats

	Availability	Confidentiality	Integrity
Hardware	Equipment is stolen or disabled, thus denying service.	An unencrypted CD-ROM or DVD is stolen.	
Software	Programs are deleted, denying access to users.	An unauthorized copy of software is made.	A working program is modified, either to cause it to fail during execution or to cause it to do some unintended task.
Data	Files are deleted, denying access to users.	An unauthorized read of data is performed. An analysis of statistical data reveals underlying data.	Existing files are modified or new files are fabricated.
Communication Lines and Networks	Messages are destroyed or deleted. Communication lines or networks are rendered unavailable.	Messages are read. The traffic pattern of messages is observed.	Messages are modified, delayed, reordered, or duplicated. False messages are fabricated.

Jazak Allah Khair!